

Table 6: Extended ablation study of sampling-aware priors and DPM fusion strategies. All models are trained on BraTS2023 and evaluated on BraTS2023 / BraTS2019.

Method	Prior				Fusion				Dice Score $\uparrow$			
	θ	A	G	D	Mul	SE	Conv	RetNet	WT	TC	ET	Avg
<i>Control Baselines</i>												
Base	–	–	–	–	–	–	–	–	90.83 / 89.44	93.05 / 90.29	92.88 / 89.64	92.25 / 89.79
θ Prior	✓	–	–	–	–	–	–	–	90.72 / 89.74	92.87 / 90.48	93.26 / 91.11	92.28 / 90.44
Aux-ViT	–	–	–	–	–	–	–	–	90.68 / 89.20	92.85 / 90.06	93.42 / 89.63	92.32 / 89.63
E2E-DVF	–	–	–	✓	–	–	–	✓	91.01 / 89.96	93.44 / 90.72	93.30 / 91.88	92.58 / 90.85
Shuffle-DVF	–	–	–	✓	–	–	–	✓	90.40 / 88.82	92.61 / 89.70	91.95 / 89.57	91.65 / 89.36
<i>Prior Comparison, fusion = Multiply</i>												
Artifact Prior	–	✓	–	–	✓	–	–	–	90.55 / 89.67	92.83 / 90.51	92.83 / 89.60	92.07 / 89.93
Gradient Prior	–	–	✓	–	✓	–	–	–	91.04 / 89.63	93.20 / 90.87	93.11 / 90.85	92.45 / 90.45
DVF Prior	–	–	–	✓	✓	–	–	–	91.04 / 90.09	93.25 / 90.96	93.28 / 91.64	92.52 / 91.03
<i>Fusion Comparison, prior = DVF</i>												
DVF + SE	–	–	–	✓	–	✓	–	–	90.87 / 89.93	93.04 / 90.89	93.04 / 91.35	92.32 / 90.72
DVF + Conv	–	–	–	✓	–	–	✓	–	91.15 / 90.26	93.36 / 91.17	93.47 / 91.62	92.66 / 90.94
SPHERE	–	–	–	✓	–	–	–	✓	91.25 / 90.25	93.67 / 91.33	93.69 / 92.24	92.87 / 91.27

θ : Direct sampling-parameter prior. **A**: Artifact Magnitude. **G**: Gradient-based Deformation Energy. **D**: Deformation Vector Field. **Aux-ViT**: Auxiliary ViT branch. **E2E-DVF**: End-to-end DVF prediction. **Shuffle-DVF**: Shuffled DVF structure.